Model Card — Gender Classification

Overview:

This project implements a gender classification system using a MobileNetV3-Small convolutional neural network trained on the UTKFace dataset.

The model predicts gender from facial images:
- 0 = Male
- 1 = Female

The model is developed using PyTorch and optimized for efficient CPU-based inference.

Dataset:

The model is trained on the UTKFace dataset, which contains facial images labelled with age, gender, and ethnicity.

The dataset contains over 20,000 facial images with labelled attributes.

The dataset includes diverse face images, but it may still contain biases related to lighting, pose, and demographics.

Model Architecture:

The model is based on MobileNetV3-Small, a lightweight convolutional neural network designed for efficient computation.

Transfer learning is used with pretrained ImageNet weights, and the final classification layer is modified for binary output.

Training Details:

- Framework: PyTorch

- Loss Function: CrossEntropyLoss

- Optimizer: AdamW

- Input Size: 224 × 224 RGB images

- Epochs: 8

- Class imbalance handled using weighted loss

Inference:

The trained model is used through a custom inference pipeline (`inference.py`), which:

- Loads the model in CPU mode
- Applies image preprocessing (resize and normalization)
- Performs prediction using the trained network
- Outputs predicted label and confidence score

Output:

The model returns:

- `label`: 0 (Male) or 1 (Female)
- `confidence`: probability score of prediction

Performance:

- Training Accuracy: 96.9%

Note: Actual performance on unseen datasets may vary due to domain differences.

Limitations:

- Assumes binary gender classification
- Performance may decrease on diverse or unseen datasets
- Sensitive to lighting, occlusion, and image quality
- May reflect biases present in training data

Future Improvements:

- Improve generalization using data augmentation
- Train on more diverse datasets
- Explore more advanced architectures
- Improve robustness to real-world variations

Conclusion:

This project demonstrates an end-to-end deep learning workflow for image classification, including model training, evaluation, and deployment-ready inference, highlighting practical implementation of efficient neural networks.